Towards the application of NeRFs and Gaussian Splitting for surface and textural representation in construction

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Abstract. This paper evaluates Neural Radiance Fields (NeRF) and Gaussian Splatting techniques, focusing on their efficacy in textural representation, surface characterization, and quality assessment of construction materials. An experiment was conducted using data collected from a sample site, where high-resolution 360-degree video data were utilized to produce NeRF and Gaussian Splatting models. Results from that experiment indicate that while NeRF effectively captures textures and surface details on nearby, well-illuminated surfaces, it struggles to reliably represent textures on distant or poorly lit surfaces. Additionally, its accuracy in reconstructing surface textures is influenced by illumination uniformity and shadow generation. Conversely, Gaussian Splatting demonstrates reduced noise and faster output generation. However, this method tends to obscure sharp edges of surface markings.

Keywords: Material conditions, Reconstruction, Neural Radian Fields, Site appearance.

1 Introduction

The digital transformation of the construction industry has progressed in the development and adoption of advanced techniques for visualizing and analyzing construction site data. Accurate and reliable methods for representing surface textures and characterizing materials are indispensable for ensuring quality and efficiency [1]. NeRF[2] and Gaussian splitting [3] are emerging techniques that offer novel approaches to capturing and rendering volumetric surfaces.

NeRF, a machine learning-based method, is renowned for its ability to generate high-fidelity 3D reconstructions with realistic texture representations, particularly in complex environments [2]. Gaussian splitting, on the other hand, offers promising potential for characterizing surface irregularities and material heterogeneity [4]. Moreover, advancements in Gaussian splatting enabled the fast and dynamic rendering of 3D environments, improving the details of the space representation and extending the real-time rendering to 4D scene capturing [5]. A comparison of the main differences of these techniques is shown in **Table 1**. The improvement of these emerging technologies over the traditional 3D representation of space, such as photogrammetry and laser scanning, enhances the surface understanding of the construction elements in the construction site. Surface understanding of the construction elements helps detect defects and understand material conditions and the state of a construction site.

This study investigates the comparative performance of NeRFs and Gaussian Splatting methods regarding surface texture representation, material differentiation, and quality control. The findings contribute to the growing body of knowledge on integrating advanced digital tools in construction management and quality assessment.

2 Background

The ability to accurately represent and understand construction material surfaces is critical for monitoring their condition, detecting defects, and ensuring quality control. NeRF and Gaussian splitting vary in their principles, expected levels of detail, and applicability, particularly when analyzing surface characteristics and general defects of materials.

2.1 NeRFs

NeRF represents surfaces by modeling volumetric scenes through neural networks. Instead of directly capturing geometric data, NeRF learns to encode scene radiance and density fields based on a set of input images[2]. This approach generates detailed and realistic reconstructions of surfaces, making it suitable for applications requiring textural conditions of construction materials. In the construction domain, Jeon et al. [6] the noel rendering capability of NeRF algorithms was utilized to recreate the realistic construction site scene for constriction progress monitoring. Moreover, Hachisuka [7] proposed a method of registering NeRF-based construction site scenes into the BIM models. Although their study utilized Instant NeRF for space reconstruction, NeRF's rendering capability was not used in the process of segmentation and registration.

The applicability of NeRF in construction scenarios is limited by its computational demands and reliance on controlled lighting conditions for input images. This is mainly because construction sites are complex and dynamic with uncontrolled illumination. In dynamic conditions like dust or variable illumination, NeRF's performance can degrade [2]. Katsatos et al. [8] performed a comparative study on 3D reconstruction methods for surface understanding in construction. In their study, they evaluated several methods, including NeRF, Patch-based MultiView Stereo(MVS), Kiss-ICP, Voxblox, and ICP Poisson Surface Reconstruction. Their results confirmed that NeRFs relatively consumed more time while methods such as Kiss–ICP produced results in less than 2 minutes. In terms of performance, Katsatos et al. [8] produced mesh surfaces and did not take advantage of the rendering capability of the methods. In terms of the meshing performance, PMVS and ICP-PSR produced relatively better results.

2.2 Gaussian Splitting

Similar to NeRF, Gaussian Splatting is a technique for 3D scene reconstruction. However, Gaussian Splatting uses Gaussian primitives to approximate light scattering and geometry, achieving high-quality rendering with efficient optimization [9]. While NeRF (Neural Radiance Fields) models the entire scene as a continuous volumetric field using neural networks and requires ray-marching for rendering, Gaussian splatting directly uses spatially distributed Gaussians, allowing faster rendering and more straightforward optimization, capturing fine surface details [10]. This can be achieved by adapting the density and distribution of Gaussians to the material's texture and geometry. It is particularly effective for surfaces with complex features like irregularities or translucency.

To take advantage of Gaussian splatting's realistic rendering capability, Cai et al. [11] proposed its implementation for developing realistic building assets in smart city management and digital twinning. Further, Gaussian splatting produced promising results in urban street surface reconstruction [12]. It can reconstruct as-built 3D models of construction sites or materials, aiding in defect detection, surface quality analysis, and material wear tracking. Its ability to integrate with photogrammetry and LiDAR enhances progress tracking, quality monitoring, and safety assessments in construction projects.

2.3 Material Conditions

While construction material surfaces can provide valuable insights into construction progress or building conditions, their digitization and recognition are often overlooked. Visual data, such as images (2D, 360, or RGBD) or laser scan data, contain textural inputs. However, this information is rarely spatially analyzed to account for location and scale. Additionally, preserving textural information as a visual layer within BIM models presents significant challenges. Recent research efforts [1,13] have developed mechanisms for representing material conditions using 2D images and 3D point clouds.

Although these methods have yielded promising results, they typically characterize surface appearances based on representative construction activities, often failing to retain their original graphical appearances. Furthermore, these representations are integrated into the IFC schema of BIM models as color-coded text information, disregarding the materials’ actual temporal appearances [14]. Preserving detailed textural appearances of buildings or construction sites offers expanded monitoring capabilities, facilitating realistic and immersive visual inspections that enhance understanding of a construction site’s current state.

Table 1. Main differences

Aspect	Neural Radiance Fields (NeRFs)	Gaussian Splitting
Purpose	Synthesizing novel views of 3D scenes via volumetric representation	Approximation of spatial distributions for efficient rendering or simulation
Core Approach	Represents a scene as a function F(x,d), predicting RGB color and density	Divides a parent Gaussian into smaller Gaussian components
Mechanism	Uses a neural network (MLP) trained via differentiable rendering	Approximates spatial distributions with a mixture of Gaussians
Applications	Photorealistic rendering, 3D reconstruction, novel view synthesis	Global illumination, light scattering, physically-based rendering
Computational Basis	Neural networks	Analytical or numerical subdivision methods
Limitations	Computationally expensive; scene-specific training required	Precision depends on splits; may increase computational overhead
Focus	Neural network-based continuous scene representation	Decomposition of spatial distributions into Gaussian components
Application to construction domain	Jeon et al. [6] the noel rendering capability of NeRF algorithms was utilized to recreate the realistic construction site scene for constriction progress monitoring. Hachisuka [7] proposed a method of registering NeRF-based construction site scenes into the BIM models. Although their study utilized Instant NeRF for space reconstruction Katsatos et al. [8]performed a comparative study on 3D reconstruction methods for surface understanding in construction. In their study, they evaluated several methods, including NeRF, Patch-based MultiView Stereo(MVS), Kiss-ICP, Voxblox, and ICP Poisson Surface Reconstruction. Their results confirmed that NeRFs relatively consumed more time while methods such as Kiss–ICP produced results in less than 2 minutes	Cai et al. [11] proposed its implementation for developing realistic building assets in smart city management and digital twinning. Further, Gaussian splatting produced promising results in urban street surface reconstruction

Although Jeon et al. [6] and Hachisuka [7] proposed NeRF-based construction site solutions, their study overlooked the rendering capability of NeRFs. Gaussian splatting directly uses spatially distributed Gaussians, allowing faster rendering and more straightforward optimization capturing fine surface details[10]. This performance can make Gaussian Splatting particularly effective for surfaces with complex features. This study examines the applicability of NeRFs and Gaussian Splats for construction surface representation.

3 Methodology

The proposed methodology evaluates the performance and applicability of Neural Radiance Fields (NeRFs) and Gaussian Splatting techniques for spatial reconstruction and the assessment of construction material conditions. This section presents a systematic approach for generating, analyzing, and validating NeRF and Gaussian Splatting models to represent material conditions at construction sites. The methodology addresses the challenges of fine-grained material representation, ensuring that the rendered models accurately depict underlying surface characteristics while minimizing the introduction of artificial artifacts.

The general methodology comprises three major sections: (1) Data Collection and Preprocessing, (2) Surface Reconstruction and Rendering, and (3) Evaluations. This research was conducted following the process outlined in **Fig. 1**.

Additionally, a region of interest is defined to identify specific surface patches in 3D space that exhibit defects, marks, cracks, or textural variations. These regions are assessed based on size, extent, and other visually identifiable characteristics. Measurements and qualitative evaluations of these regions are documented as ground truth for comparison with the rendering results produced by NeRFs and Gaussian Splatting techniques.

3.1 Surface reconstruction and rendering

The preprocessed data generates 3D scene reconstructions and rendering using NeRFs or Gaussian splats. The process involves generating spatial data, training NeRF and Gaussian Splats models, defining rendering scenes (camera trajectory), and performing the rendering process.

The spatial data can be generated following a Structure from Motion (SfM) pipeline where images are ordered, features extracted and matched, and camera pause and geometric 3D space can be estimated. For NeRFs, the approach optimizes a volumetric scene representation by minimizing the reconstruction error between the observed images and the rendered outputs. The optimization process relies on different rendering techniques to model the radiance fields accurately.

In the case of Gaussian splats, the method involves fitting Gaussian distributions to represent the spatial and photometric properties of the scene. These splats are positioned in the 3D space to reconstruct surfaces with varying levels of granularity. The Gaussian splats are designed to preserve macroscopic and microscopic details, enabling the evaluation of surface irregularities and potential defects. The same dataset and preprocessing steps are used for both NeRFs and Gaussian splats to ensure a fair comparison.

3.2 Evaluation

Once the 3D scene is generated and rendered, the next step involves assessing their ability to represent surface characteristics accurately. This evaluation focuses on grain size distribution, texture, and defects or irregularities. The analysis is performed by comparing the rendered models with ground truth data. The metrics used in this study include spatial resolution (around a selected region of interest), rendering quality in terms of the defect measurement, and the level of detail preserved.

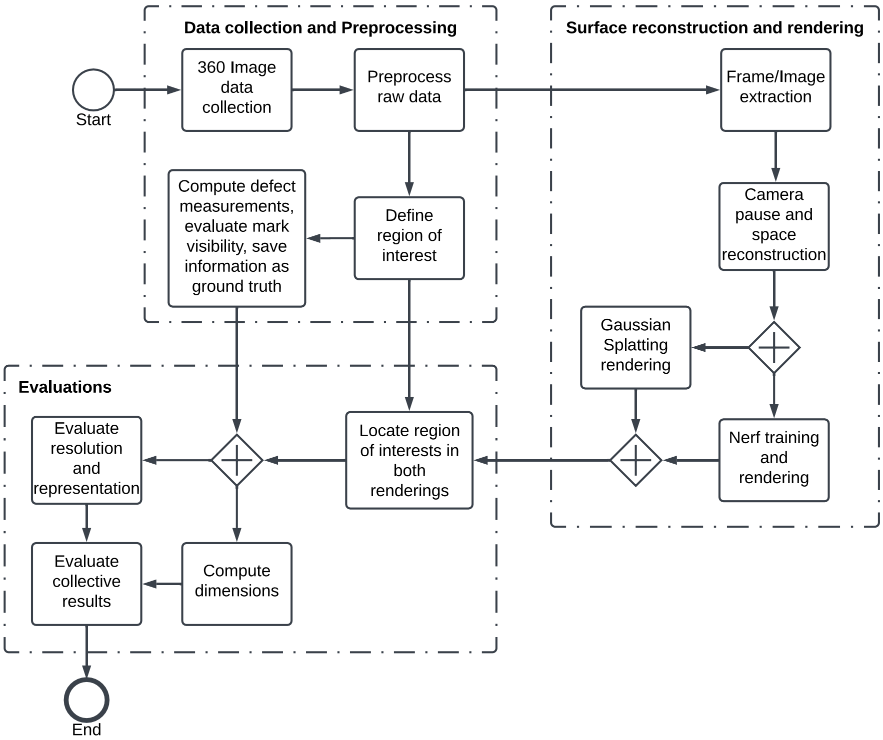


Fig. 1. Overview of the main elements of the proposed methodology

4 Case Study

The experiment was conducted in a research laboratory featuring exposed concrete ceilings and floors. The ceiling exhibited visible wires, markings, and casting irregularities distributed randomly. The floor included masking tapes (blue lines) in various colors, sizes, and orientations. Additionally, the floor featured two distinct textures: carpeted and plain concrete screed. Illumination in the room was provided by strip fluorescent lights evenly positioned across the space, facing downward. This arrangement resulted in relatively higher illumination levels on the floor and lower levels near the ceiling. Representative images of the sample site are shown in **Fig. 2**(a).

4.1 Data collection and preprocessing

360 video data was collected from the sample site using the Insta360 camera[15]. The raw 360 footage was processed using Insta360studio[15] to generate a 360 video. The 360 video was used to create planar images. These planar images were downsampled and processed further using COLMAP[16,17] to perform the SfM pipeline. The resulting textured 3D information (see **Fig. 2**(b) and (c)) was used to train the NeRF and Gaussian Splat models. The detailed characteristics of data inputs and outputs in this experiment are summarized in **Table 2**. The process is conducted using Web-based platform[18] with available GPU machine model L40S, 48GB memory and 16CPU.

The ground truth was selected from the actual-size scene to evaluate the accurate representation of the material conditions in the rendering using NeRF and Gaussian Splats. The selected ground truth points include the variations in the carpet and concrete screed surface, the complete representation of the blue marking across the floor, and the complete representation of the cables and formwork markings on the ceiling.

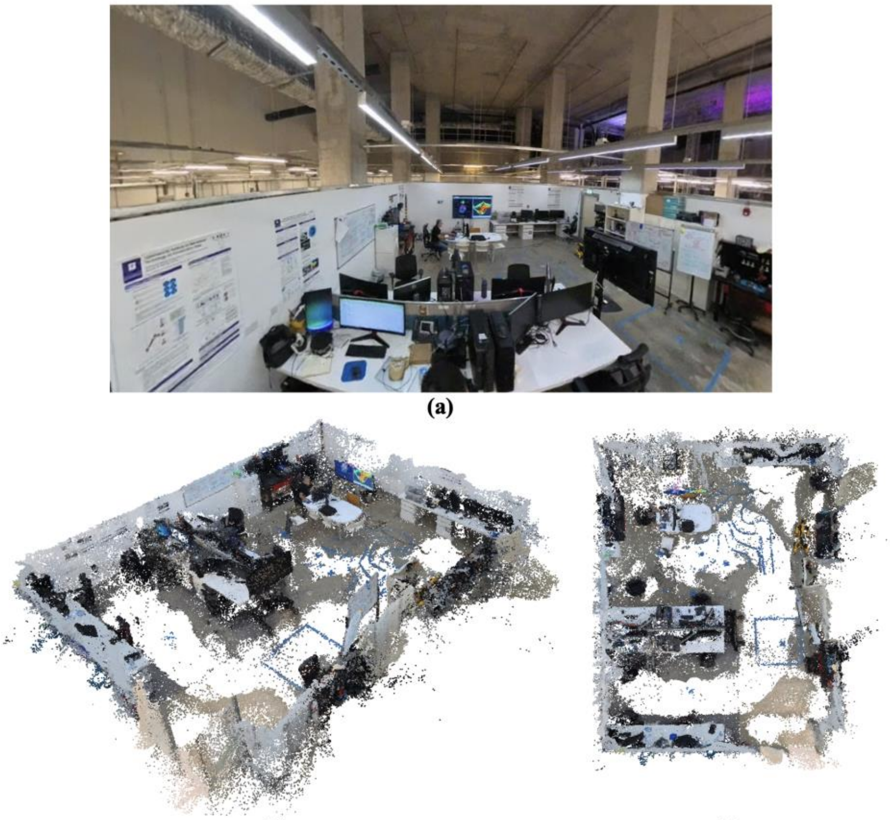
Table 2. Data preprocessing

Data/Process	Size and characteristics of data	Processed using
360 video	60 seconds, 466MB 3840x1920 resolution, 60fps	Insta360 studio
Planar images	5304 images, each 960 x 960 resolution,	NeRF Studio – Data processing
Downscaling	All of the planar images (three scales, 2,4 and 8)	NeRF Studio – Data processing
Feature matching and 3D space	Point cloud (1.1mil points)	COLMAP

4.2 Surface reconstruction and rendering

Surface reconstruction is conducted using NeRF Studio [19] and the outputs of the COLMAP process, which contains textural and spatial information. The method of the NeRF rendering starts by training a Nerfacto model[20] on Nerfstudio. Nerfacto is the defacto NeRF model built by NerfStudio, combining several published methods to perform camera pose refinements, piecewise and proposal sampling, scene contraction, and hash encoding. Once the model is trained on the collected data, PortViewer on Nerfstudio defines the preferred camera path and rendering.

The COLMAP result is again utilized for rendering using Gaussian splatting. Gaussian Splatting is implemented in this research using Splatfacto[21], a method that is available in the Nerfstudio methods library. Splatfacto is built based on [3] and several other methodologies. The process details are presented in **Table 3**.



(a)

(b) (c)

Fig. 2. (a)Sample site characteristics including the General site appearance, the Ceiling, the floor with blue marks, carpeted and screed areas SfM reconstruction of the sample space (b) general 3D view, (c) top view with visible floor markings

5 Results and discussions

The developed model successfully rendered visual representations of ground truth elements, including floor markings, variations between carpet and screed textures, and ceiling markings. The rendered views/outputs were evaluated by comparing them against selected textural features, which served as predefined evaluation points. These comparisons aimed to assess the accuracy and fidelity of the model in replicating the original textural details.

As shown in **Fig. 3**, the floor markings (e.g., blue lines) are reasonably visible, though some edge dilation is apparent. These markings were generally well-represented during the 3D reconstruction phase. From **Fig. 3**, points A1, A2 and A3 are white spots on the floor that exist physically. While A2 is closely aligned with A3 in the rendered output, A1 appears blurred. Similarly, the mat contains white marks (see B1, B2, and B3 on **Fig. 3**).

Table 3. General surface reconstruction process overview

Process	Processed using	Time taken to process (min)	Resulting output
COLMAP	COLMAP library within the NeRF studio	74	Point cloud, texture
NeRF training	NeRF studio, Nerfacto	120	Deep learning model, trained to produce a rendered view of a selected camera frame
NeRF rendering (30fps)	PortViewer	141	Rendered video
Gaussian splats	NeRF studio, Splatfacto	43	Model

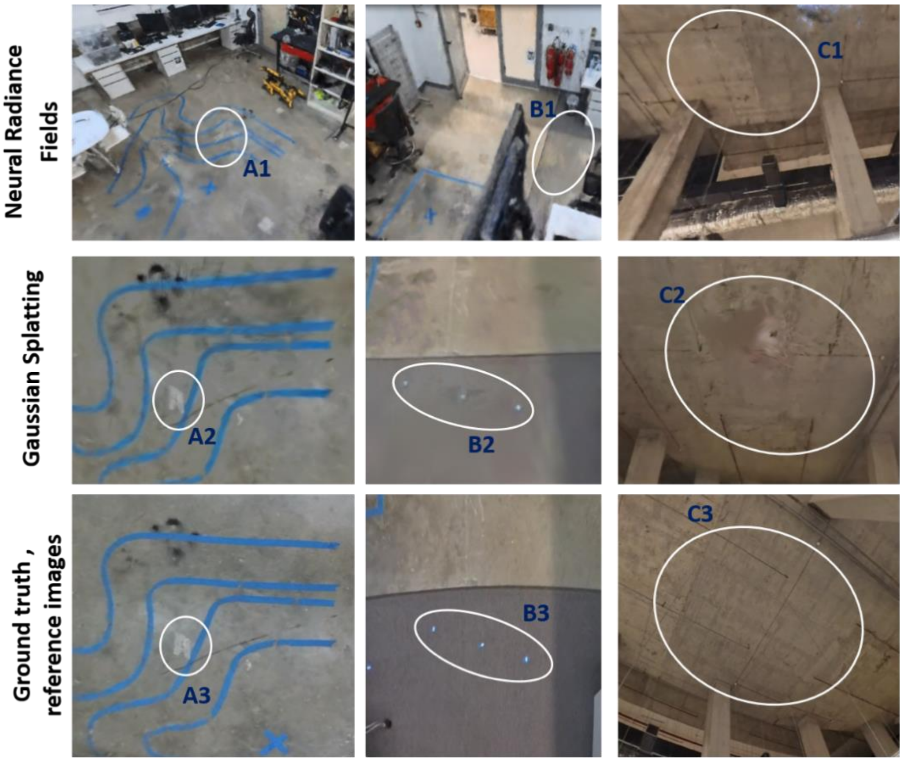


Fig. 3. NeRF and Gaussian Splatting outputs

While B2 and B3 have strong similarity, demonstrating the performance of Gaussian Splats, the white marks at B1 are blurred and partially obscured. Regarding the floor surfaces, the floor mat is distinguishable from the surrounding screed texture; however, in certain shadowed areas, NeRF replaced the floor mat texture with one resembling concrete screed. This inconsistency highlights the influence of shadows and uneven illumination on NeRF's output accuracy. The ceiling was also adequately depicted; however, some markings (see C3 on **Fig. 3**) depict notable features such as markings from casting and wires. While these features are generally visible in C1, C2 includes extraneous splats that obscure the original details.

Gaussian Splats yielded smoother surface representations with reduced noise from the SfM process. However, its effectiveness depends heavily on the camera's proximity and orientation to the surface. In areas like region C2, Gaussian Splats introduced artificial marks and openings that obscured actual ceiling lines. This tradeoff between noise reduction and detail preservation limits its applicability for precise site inspections, defect detection, and progress monitoring. Notably, Gaussian Splatting requires less computational time, as shown in **Table 3**, with NeRF training being three times more time-intensive and computationally demanding for rendering. While NeRF achieves higher detail fidelity, its significant resource demands may limit practical applications, especially in dynamic or large-scale environments.

6 Conclusion

This study evaluated NeRF and Gaussian Splatting techniques for their ability to represent surface textures, characterize materials, and assess quality in construction environments. NeRF demonstrated strong capabilities in capturing detailed textures and surface features on well-illuminated and nearby surfaces, but its performance diminished for distant or poorly lit areas due to sensitivity to illumination and shadow effects. Moreover, NeRF required significantly more computational resources and time, making it less practical for dynamic or large-scale applications. On the other hand, Gaussian Splatting provided faster processing, reduced noise, and smoother surface representations, but it occasionally obscured sharp edges and fine details, especially in areas requiring high-fidelity texture replication. This method's effectiveness was influenced by the proximity and orientation of the camera to the surface, highlighting the importance of optimized data collection. While NeRF is better suited for detailed inspections under controlled conditions, Gaussian Splatting offers an efficient alternative for broader assessments where speed and noise reduction are prioritized. The findings underscore the complementary nature of these technologies, suggesting potential benefits in integrating their strengths to enhance construction site monitoring, defect detection, and progress evaluation. Future research should explore hybrid approaches to address their respective limitations and maximize their applicability in real-world scenarios.

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